

Behavioral Anomaly Detection in Online Chess:
Unsupervised Identification of Engine-Assisted Play

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Background

Engine-assisted play leaves measurable statistical fingerprints: move quality, game length efficiency, and rating dynamics all deviate from population norms. No reliable ground-truth labels exist — platforms cannot publicly confirm individual cases. We apply unsupervised anomaly detection to player-level behavioral aggregates to flag statistically unusual players for human review.

Research Question

Among players in the same Elo rating band, whose observed behavioral statistics are statistically inconsistent with the bulk of that population?

Dataset

- Lichess July 2016 — 6.25M games; 500k sampled
- Rapid + classical only — engine use requires thinking time
- 27,909 unique players → 17,909 after quality filter (≥ 5 games)
- Player-level split: 70% train / 15% val / 15% test

Permutation Feature Importance

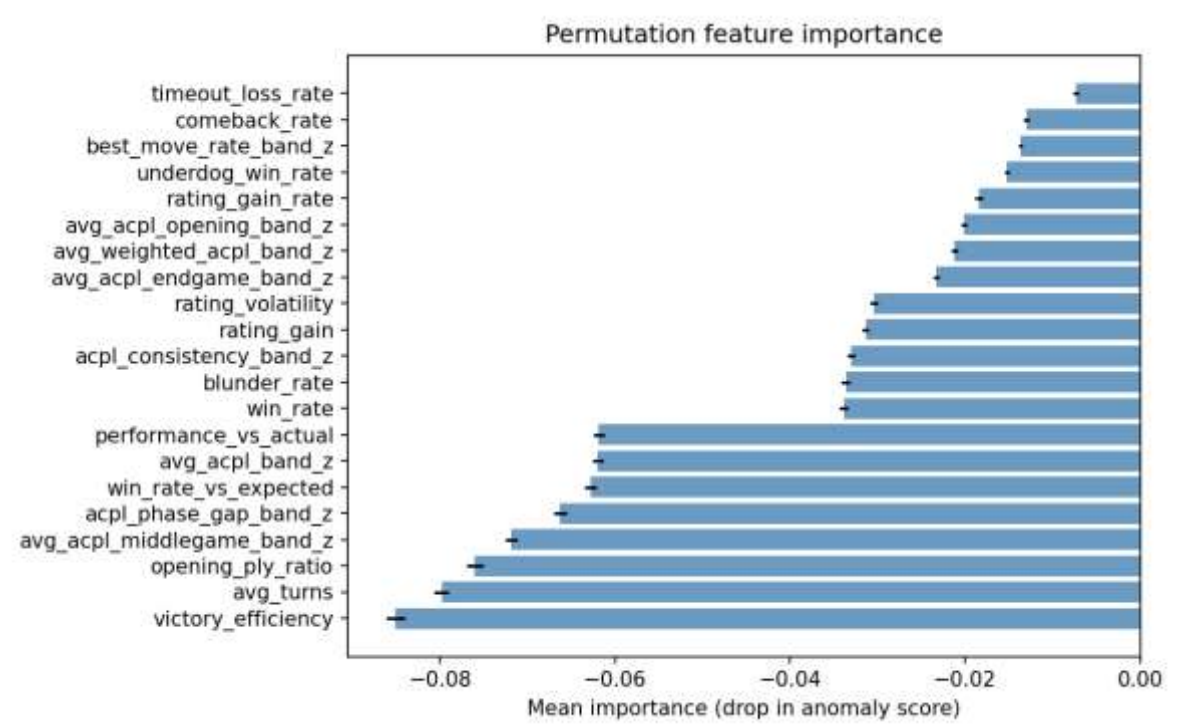


Fig 1. Permutation importance — more negative = stronger anomaly signal.

Key Statistics

27,909

Players Analyzed

17,909

Players Filtered

21

Features Engineered

312

Players Flagged

Feature Engineering

Behavioral (8 features)

8 statistics — *win_rate_vs_expected*, *victory_efficiency*, *opening_ply_ratio*, *rating_volatility* — capturing playing style vs. Elo-predicted outcomes.

Stockfish ACPL (13 features)

13 signals: ACPL by phase, *blunder_rate*, *best_move_rate*, *acpl_phase_gap*. Band-normalised within 9 Elo tiers; z-scores refit on training split only.

Models & Ensemble

Model	Type
ZScore Baseline	Statistical threshold
Isolation Forest	Tree ensemble
LOF ★ BEST	Density-based
One-Class SVM	Kernel boundary
Autoencoder	Neural reconstruction
HDBSCAN	Clustering

Ensemble Design

Top 3 by CV AUC: LOF (0.9585), Autoencoder (0.8925), OC-SVM (0.8769). Isolation Forest (0.7598) excluded — high redundancy with LOF. HDBSCAN (AUC 0.52) excluded — near-random, no stable cluster structure. Majority vote (≥ 2/3) → 312 flags (2.5%); not a preset threshold.

Validation

Synthetic injection at three levels: *engine_perfect* (→ 99th percentile), *subtle* (±1.5 SD on ~1/3 features), *realistic_cheater* (low ACPL only). 5-fold CV on training split; val/test held out for final evaluation.

Model Agreement Matrix

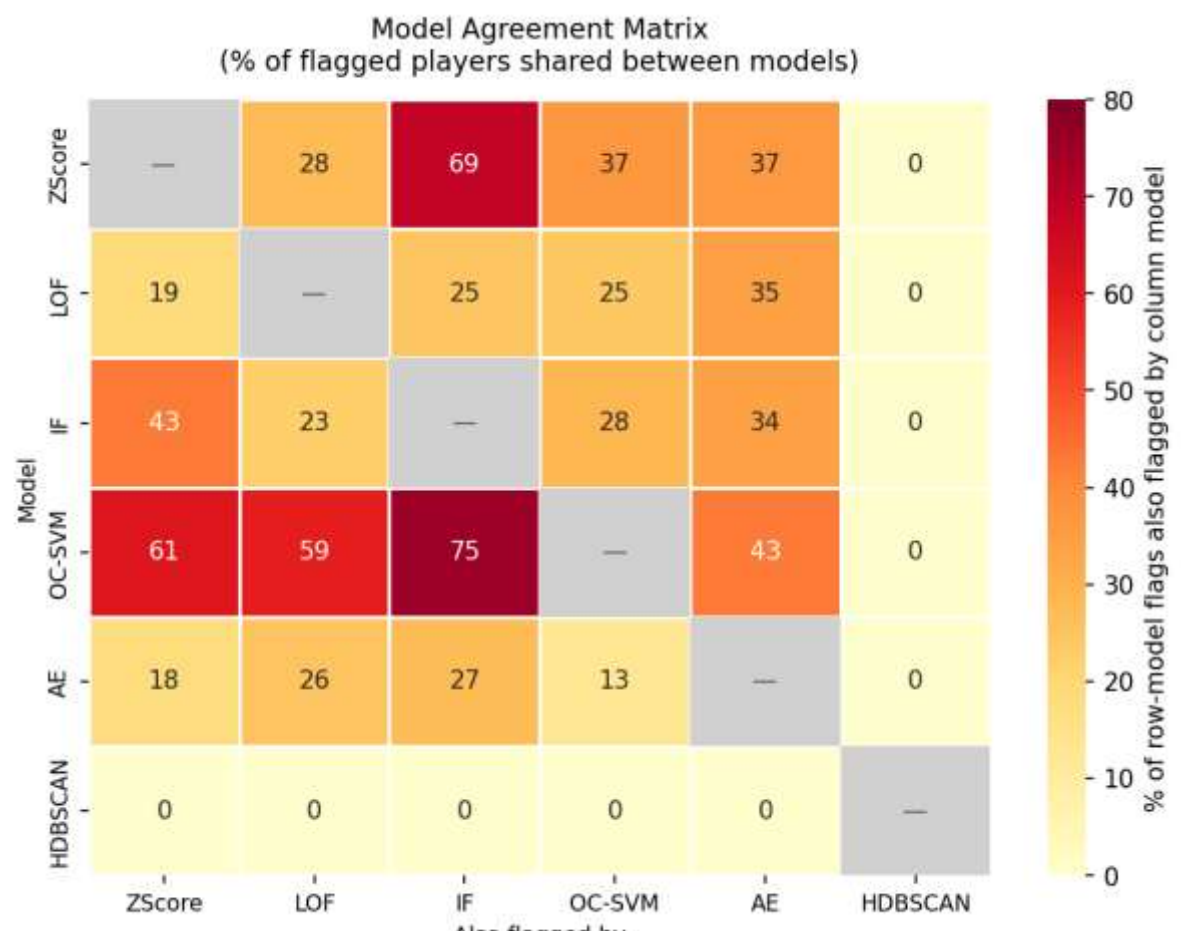


Fig 4. Pairwise flag overlap — low overlap confirms ensemble complementarity.

Results

Model Performance (5-Fold CV + Subtle Injection Test Set)

Model	CV ROC-AUC	Test AUC
LOF ★	0.9585 ± 0.0305	0.9706
Autoencoder	0.8925 ± 0.0231	0.9069
One-Class SVM	0.8769 ± 0.0210	0.8895
ZScore Baseline	0.7802 ± 0.0314	0.8053
Isolation Forest	0.7598 ± 0.0162	0.7979
HDBSCAN	0.5242 ± 0.0541	0.6250

ROC Curves — Subtle Injection Test Set

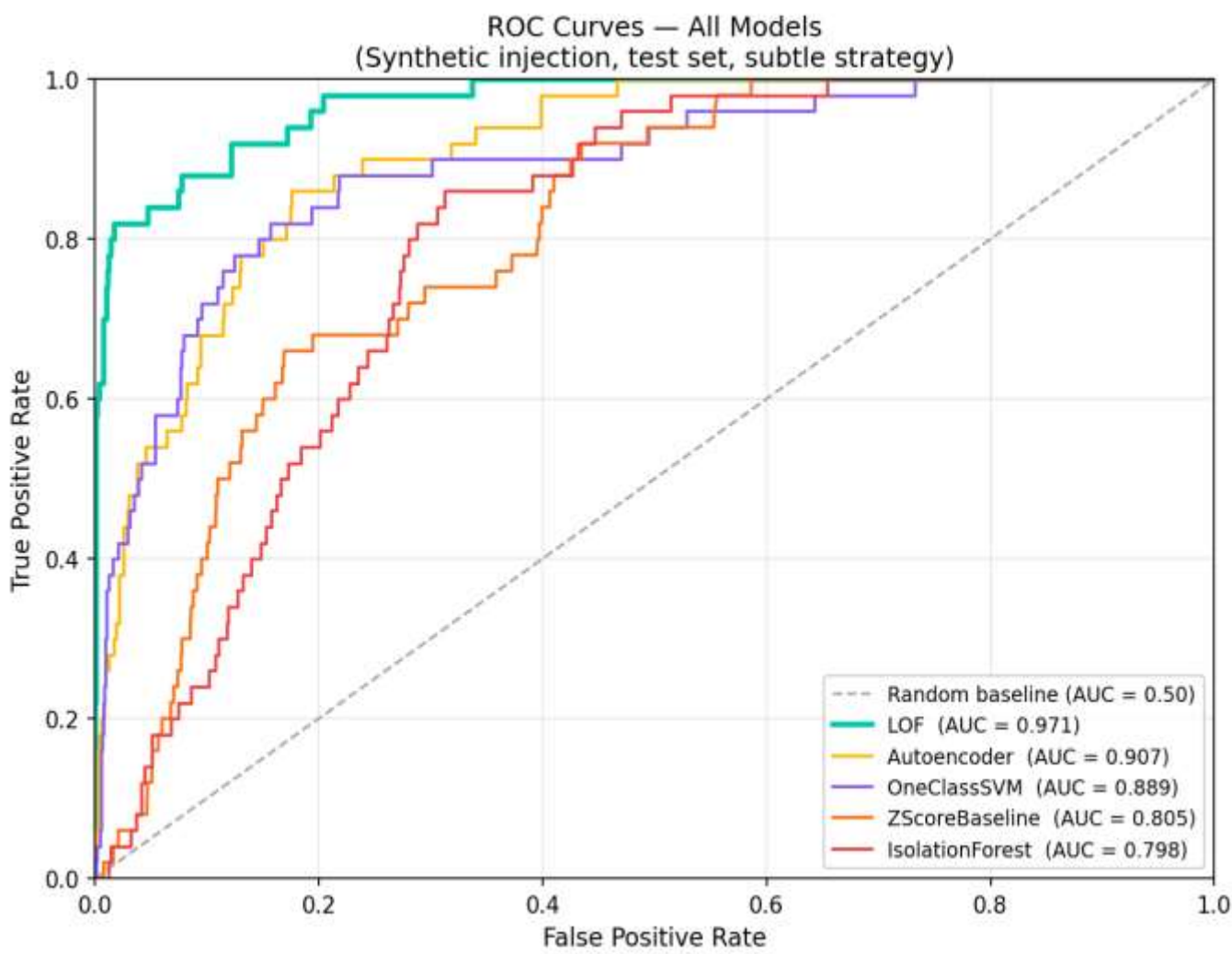


Fig 2. ROC curves for all models — subtle-injection test set.

Feature Space — UMAP Projection

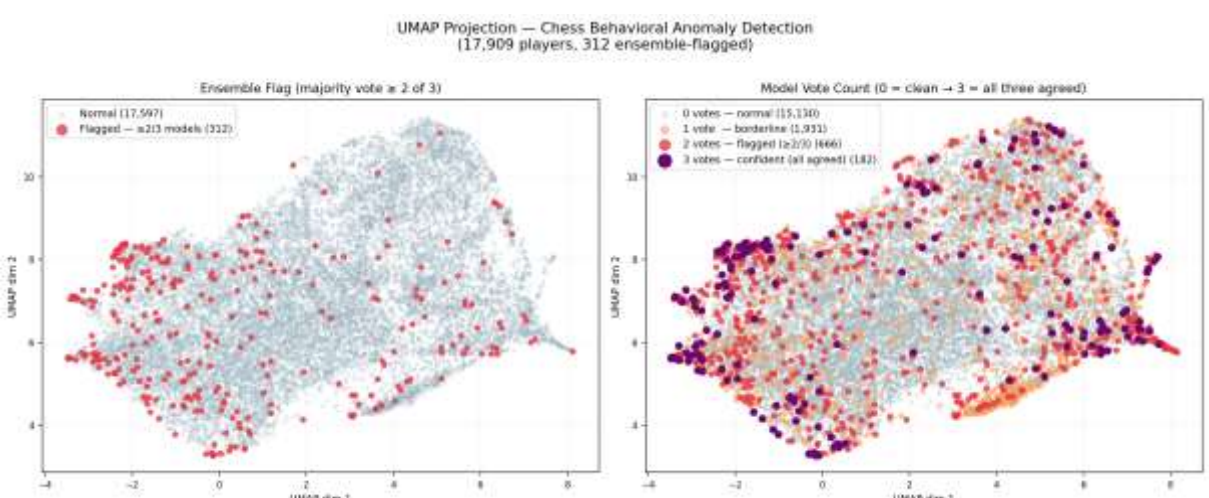


Fig 3. UMAP of 21-feature player space; colored by LOF anomaly score.

Conclusion

LOF achieved CV ROC-AUC 0.9585 ± 0.0305 and Test AUC 0.9706, outperforming all baselines. The ensemble captures complementary patterns — only 26–35% flag overlap. 312 of 17,909 players (1.7%) flagged via ≥ 2/3 vote — not a preset threshold; triage only.

Limitations & Future Work

No ground-truth labels; all metrics are synthetic-injection based. Lichess 2016 lacks clock data; *move_time_cv* could not be computed. Future: 2017+ data with clocks, SHAP explanations, active learning.

Acknowledgements — We thank Prof. Turilli for supervision. Data: Lichess.org (CC licence).